

Learning-to-Defer with Active Human Feedback

News topic classification on AG News with a simulated expert and adaptive user study

1 Introduction and project objective

This project studies selective human-AI decision making for four-class news classification. Instead of forcing the classifier to answer every article, the system can defer a decision when a human expert is expected to be more reliable. The work is organized as a progression: a baseline AI classifier (Task 1), a deliberately imperfect simulated expert (Task 2), a staged learning-to-defer system (Task 3), a limited-budget active-learning experiment for expert competence discovery (Task 4), and a participant-facing adaptive feasibility pilot that updates an individual human-competence model from responses online.

All model experiments use AG News: 120,000 training articles and 7,600 test articles across World, Sports, Business and Sci/Tech. The held-out test split is used only for model evaluation. The final web interface separates the research study from the technical demonstrations: the landing page exposes only the project report and the user study as primary actions, while Tasks 1-4 and the live classifier are kept as additional technical resources.

Table 1: Project structure and the role of each component.

Component	Purpose	Main design choice
Baseline AI	Classify every article	TF-IDF + multinomial logistic regression
Simulated expert	Create complementary expertise	Class-dependent competence
Learning to defer	Choose AI or expert	Compare AI confidence with estimated expert competence
Offline active learning	Reduce expert-query cost	Query near the defer/answer boundary
Adaptive user study	Feasibility pilot for real-human feedback	10 adaptive trials from a 20-article pool

2 Task 1: Baseline classifier

2.1 Method and motivation

The baseline uses TF-IDF text features followed by multinomial logistic regression, trained on all 120,000 labelled training articles. The vectorizer lowercases text, removes English stop words, uses unigrams and bigrams, applies sublinear term-frequency scaling, limits the vocabulary to 100,000 features, and uses minimum document frequency 2 and maximum document frequency 0.95. The logistic-regression model uses the lbfgs solver and random state 42.

This model was chosen because it is fast, interpretable and competitive for topic classification, while also producing class probabilities. Those probabilities are essential later: the maximum probability is used as AI confidence, and the complete four-probability vector becomes the input to the competence model. Using one simple baseline throughout also isolates the effect of deferral and active learning from changes in the underlying classifier.

2.2 Results

The baseline reaches 92.24% test accuracy. Sports is the strongest class, while Business and Sci/Tech are more difficult and show more overlap.

Table 2: Baseline test performance (overall accuracy 92.24%).

Class	Precision	Recall	F1-score	Support
World	0.937	0.911	0.924	1,900
Sports	0.956	0.982	0.969	1,900
Business	0.898	0.890	0.894	1,900

Sci/Tech	0.898	0.906	0.902	1,900
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3 Task 2: Simulated expert

3.1 Design and justification

The simulated expert is intentionally complementary rather than globally superior. It is a Business-news specialist with designed correctness probabilities of 97% for Business, 86% for World, 84% for Sci/Tech and 70% for Sports. When wrong, it chooses one of the other three classes. Correctness is generated deterministically from the article text and a fixed seed, so the same article always receives the same expert answer.

Class-dependent competence creates meaningful regions where human expertise can help and regions where deferral is harmful. This is important because a learning-to-defer system is only interesting when the team members have different strengths. The expert is therefore weaker than the AI overall but markedly stronger on Business, where the AI is comparatively weaker.

Table 3: Simulated expert accuracy (overall 84.26%).

Class	Designed accuracy	Measured accuracy	Test samples
World	86.0%	85.9%	1,900
Sports	70.0%	71.0%	1,900
Business	97.0%	97.5%	1,900
Sci/Tech	84.0%	82.6%	1,900

4 Task 3: Learning to defer

4.1 Staged competence model

A logistic-regression competence model predicts the probability that the expert will be correct. Its four inputs are the baseline classifier's class probabilities; the target is 1 when the expert is correct and 0 otherwise. At inference time, the system defers when estimated expert competence is greater than AI confidence. The deferral margin is fixed at 0, so the decision rule treats the two team members symmetrically.

A staged approach was preferred to a joint learning-to-defer loss because it is transparent, reuses the Task 1 classifier unchanged, and can be retrained cheaply when only a small set of expert queries is available in Task 4. Joint surrogate-loss approaches are more theoretically integrated, but the staged design makes the source of each decision easier to inspect and explain [2].

4.2 Team performance and deferral quality

The combined team reaches 94.84% test accuracy, improving on the AI alone (92.24%) and the expert alone (84.26%). The system defers 37.54% of test articles (2,853 of 7,600). There are 424 beneficial deferrals where the expert corrects an AI mistake and 226 harmful deferrals where the AI was right but the expert was wrong. The oracle upper bound, which always chooses the correct team member when either is correct, is 99.17%.

Table 4: Learning-to-defer results on the held-out test set.

Metric	Value
AI-only accuracy	92.24%
Expert-only accuracy	84.26%
Human-AI team accuracy	94.84%
Oracle upper bound	99.17%
Deferral rate	37.54%
Accuracy on deferred articles	90.08%
Accuracy on non-deferred articles	97.70%

Beneficial / harmful deferrals	424 / 226
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Table 5: Per-class accuracy and deferral behaviour.

Class	AI	Expert	Team	Deferral rate
World	91.1%	85.9%	94.0%	27.7%
Sports	98.2%	71.0%	96.7%	11.2%
Business	89.0%	97.5%	95.6%	73.4%
Sci/Tech	90.6%	82.6%	93.1%	37.8%

5 Task 4: Active learning for expert competence discovery

5.1 Query setting and design choices

Task 4 removes full access to expert labels. The baseline classifier still uses all labelled training data, but expert correctness is observed only for explicitly queried articles. Each run begins with 40 queries, stratified over the four AI-predicted classes (approximately 10 per predicted class). Forty was chosen as a small but stable initialization: it exposes the competence model to all four regions and reduces the risk that the queried correctness targets contain only one class, in which case logistic regression cannot be fit.

After initialization, active querying selects unqueried articles with the smallest absolute difference between estimated expert competence and AI confidence. These are the examples closest to the defer/answer boundary, where an additional expert observation is most relevant to the final decision. This is an uncertainty-sampling strategy focused on the decision boundary rather than on the already-supervised topic classification task [3]. Random querying is used as the control. Budgets are 50, 100, 250, 500, 1,000 and 2,000 queries, and every configuration is repeated with five random seeds. The offline experiment queries and retrains in batches between these budget checkpoints; it is therefore different from the participant study, which updates the human-competence state and selects the next item after every individual response.

5.2 Results

The observed mean results suggest that active querying is most useful at low budgets. With 50 expert queries, the mean team accuracy is 94.66% for active querying and 94.23% for random querying. The difference becomes smaller as the budget grows; at 2,000 queries the corresponding means are 94.97% and 94.90%. In Table 6, the “ $\pm$ ” values denote the standard deviation across five random seeds. These descriptive results indicate a promising low-budget pattern, but five-seed variation alone is not a statistical significance test and should not be interpreted as establishing that active querying is universally superior.

Table 6: Mean team accuracy $\pm$ standard deviation across five seeds, with competence ROC-AUC shown descriptively.

Budget	Random team acc.	Active team acc.	Random AUC	Active AUC
50	94.23% $\pm$ 0.52	94.66% $\pm$ 0.29	0.626	0.639
100	94.49% $\pm$ 0.05	94.70% $\pm$ 0.36	0.656	0.642
250	94.58% $\pm$ 0.12	94.71% $\pm$ 0.33	0.676	0.659
500	94.70% $\pm$ 0.23	94.86% $\pm$ 0.25	0.681	0.675
1,000	94.87% $\pm$ 0.11	94.96% $\pm$ 0.17	0.681	0.678
2,000	94.90% $\pm$ 0.07	94.97% $\pm$ 0.06	0.682	0.680

6 Adaptive participant user study

6.1 Participant flow and interface separation

The final interface provides a self-contained feasibility-pilot flow that does not require the participant to understand the technical pipeline. From the Project 3 landing page, the two primary actions are “Download PDF” and “Enter User

Study”. The technical walkthrough is intentionally secondary. The participant sees an instruction page, starts a fresh study session, classifies 10 articles one at a time, and ends on a completion screen with an anonymous study identifier and a route back to the project overview. Before collecting real participant data, the server-side trial-validation and duplicate-submission safeguards described in Section 6.6 should be added.

The participant-facing study is separate from the technical Human Expert demonstration. Correctness, ground truth, AI prediction, AI confidence, estimated competence and defer/answer decisions are hidden during the 10 trials. This was a deliberate methodological choice: revealing the correct answer after each trial would teach the participant the dataset’s category boundaries and could change later responses. The system still uses the hidden ground truth internally to update the participant model.

6.2 Why 10 trials from a 20-article pool?

The prepared candidate pool contains 20 training articles taken from the final active-learning query sequence. They are deliberately concentrated near the original AI-expert deferral boundary, where collaboration decisions are informative. Each participant sees only 10 trials. Keeping the pool larger than the session length gives the adaptive selector genuine choice and allows different participants to receive different sequences, while limiting the session to 10 items reduces fatigue and keeps the study practical. Because this pool is deliberately selected rather than sampled representatively from the held-out test set, performance observed in the participant study must not be interpreted as representative test accuracy or directly compared with the held-out results in Tables 2-5.

6.3 Online learning of participant competence

The user study learns a participant-specific competence estimate separately for World, Sports, Business and Sci/Tech. Each category starts with a Beta(2,2) prior, whose mean is 0.5. After a response, the hidden ground truth determines whether that response was correct. For the true category, a correct response increments alpha and an incorrect response increments beta. After c correct and w incorrect observations in a category, the estimate is $(2 + c) / (4 + c + w)$. The Beta(2,2) prior was chosen instead of an unregularized empirical rate so that a single early response cannot push competence immediately to 0% or 100%. For example, one correct response moves 0.50 to 0.60; two correct responses move it to approximately 0.667.

6.4 Adaptive selection of the next article

The first article is a random seed because no participant-specific observations exist yet. Every later article is selected only after the previous human response has updated the competence model. Candidate articles already shown are excluded. Because the true class of an unshown article must not be used to choose it, the selector uses the AI-predicted class to look up the current human competence estimate for that region.

For each remaining candidate, the implementation combines four normalized signals into a fixed heuristic score: 45% human-competence uncertainty, 30% closeness between learned human competence and AI confidence, 15% AI uncertainty, and a 10% bonus for a predicted category with no participant observations yet. The weights are fixed before data collection rather than tuned on participant outcomes. They prioritize learning where the human model is uncertain, then focus on the decision boundary, while still encouraging category coverage. The participant never sees the score or why a particular article was chosen.

Table 7: User-study design choices and motivation.

Study choice	Final implementation	Why
Candidate pool	20 boundary-focused articles	Preserves informative alternatives for adaptive selection
Trials per participant	10	Limits fatigue while allowing repeated online updates
Initial human model	Beta(2,2) per category	Regularized 50% prior; avoids extreme estimates after one trial
Next-item selection	Adaptive after each response	Makes later trials depend on what was learned from the human
Correctness feedback	Hidden	Avoids teaching category boundaries and biasing later responses
Primary study data	JSON per anonymous study ID	Supports partial saves, timing analysis and reproducibility
Evaluation scope	Feasibility pilot	Demonstrates adaptive collection/competence learning; not a collaboration-gain estimate

6.5 Data recorded

A new pseudorandom study ID is created whenever “Start User Study” is pressed. The study form does not ask for a participant name or other direct identifier. Results are saved locally after every response, not only at the end, so incomplete sessions are retained. Each JSON file stores timestamps, selected article indices, human labels, hidden ground truth, correctness, response time, AI prediction and confidence, competence before and after the update, and the adaptive-selection metadata. This makes it possible to reconstruct how the human model and query sequence evolved during each session. Existing local files created during development and debugging should be archived separately and excluded from the future participant-data directory before formal collection begins, so that test sessions cannot be mixed with study observations.

6.6 Scope of the pilot and pre-collection safeguards

Before participant data collection, each rendered trial should also carry a server-generated, server-validated trial identifier tied to the active study session and expected trial index. A submitted trial identifier should be accepted at most once; repeated POSTs caused by double-clicking, refresh, browser resubmission or back-navigation should be rejected or treated idempotently. This protects response counts, competence updates and adaptive-selection state from accidental duplicate submissions.

The present participant study is a feasibility pilot. Its purpose is to verify that participant labels can be collected, category-specific competence can be updated online, and later article selection can react to those updates. It does not currently evaluate a participant-specific answer/defer policy, and it should therefore not be used to claim real-human collaboration gains over AI-only or human-only baselines. A study intended to estimate collaboration gains would need a separate evaluation phase on a clearly defined evaluation sample, ideally distinct from the adaptively selected training/query pool. Any deliberately selected evaluation sample should be reported as such and kept separate from representative held-out test performance.

7 Interface, live classifier and reproducibility

The Django interface has three clearly separated roles. The landing page is a minimal entry point for the report and user study. Additional Resources contains only the technical progression (Tasks 1-4) and the Live Classifier. The Live Classifier accepts arbitrary article text and displays the baseline prediction, class probabilities, AI confidence, estimated expert competence and the resulting answer-or-defer decision. It is a demonstration of the Task 3 policy and is not part of the participant experiment.

The model artifacts and aggregate metrics are stored in `learning_to_defer/artifacts`. During development, local user-study JSON files are written under `learning_to_defer/artifacts/user_study_results/`. Before formal participant collection, existing trial/debug files should be moved to a separate archive and a clean study-data directory should be used for new sessions. The principal experiment artifacts can be regenerated in the following order:

```
python -m learning_to_defer.services.train_baseline
python -m learning_to_defer.services.evaluate_expert
python -m learning_to_defer.services.evaluate_deferral
python -m learning_to_defer.services.evaluate_active_learning
python -m learning_to_defer.services.prepare_human_expert_pool
python manage.py runserver
```

8 Limitations and interpretation

- The competence model in Tasks 3-4 uses only four AI probability features; richer text or representation features may identify competence regions more precisely.
- The zero deferral margin is deliberately simple but still allows harmful deferrals, especially for Sports where the AI is already very strong.
- The participant study is explicitly a feasibility pilot with a 20-article boundary-focused pool and 10 trials per session. It demonstrates adaptive human modelling and data collection, but the deliberately selected pool is not representative of the held-out AG News test distribution. Accordingly, participant-study accuracy should not be treated as representative test performance or as evidence of a real-human collaboration gain. Measuring collaboration gains would require a separate evaluation phase and a clearly distinguished evaluation sample.
- Online participant competence is updated using hidden dataset ground truth. That is valid for this controlled study, but a deployment without immediate verified outcomes would require delayed labels or another feedback source.

- The adaptive user-study score is a fixed heuristic rather than an optimized acquisition function. Its weights should be validated or tuned in a larger study before making strong claims about optimal query efficiency.
- The current prototype should be hardened before real participant collection with a server-validated trial identifier and duplicate-submission protection, and development/debug session files should be physically separated from the clean study dataset.

9 Conclusion

The project demonstrates a complete path from classification to adaptive human-AI collaboration. A 92.24% baseline classifier and an 84.26% complementary expert combine to reach 94.84% team accuracy when the system learns when to defer. Offline active learning shows that boundary-focused expert queries are particularly useful when the annotation budget is small and that near-full-supervision performance is reachable with fewer than 2,000 expert queries.

The final user study extends the project from a fully simulated expert toward a real-human feasibility pilot: each participant response is stored, used to update category-specific competence online, and then influences which article is shown next. At the same time, the participant interface hides correctness and system-internal information to reduce sequential bias. The current study demonstrates adaptive data collection and competence learning; it does not yet evaluate a participant-specific answer/defer policy or establish real-human collaboration gains. Those questions require a separate, clearly defined evaluation phase.

References

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- [2] H. Mozannar and D. Sontag, "Consistent Estimators for Learning to Defer to an Expert," *Proceedings of the 37th International Conference on Machine Learning*, PMLR 119, pp. 7076-7087, 2020.
- [3] B. Settles, "Active Learning Literature Survey," *Computer Sciences Technical Report 1648*, University of Wisconsin-Madison, 2009.